

Basic sound setup

About **\$450** with a borrowed calibrator; **\$700** with our own.

Use our computer and neural recorder. This is the simplest proposed setup for selected tones below 20 kHz. The microphone checks what reaches the ear.

Computer → **USB interface** → **powered speaker**

Microphone near ear → **interface** → **sound recording**

Parts to buy

Blue names are buying links. USD asking-price snapshot: October 9, 2026.

Job	Part / seller	Price
Play and record	Behringer UMC204HD – eBay / Sweetwater, new	97.98
Make the sound	One JBL 305P MkII powered speaker – Reverb, new/open box	122.39
Measure sound	Dayton EMM-6 microphone – eBay / Parts Express, new	66.93
Mic cable	Hosa HMIC-010 XLR cable – Sweetwater	17.99
Audio cables	Two Hosa HSS-005 TRS cables , \$14.95 each	29.90
Hold the mic	Gator RI-MICTP-FBM stand – Sweetwater	34.99
Parts	Borrow the sound-level calibrator	370.18
Optional purchase	REED R8090 calibrator – DigiKey / REED, new	219.00

Working totals: \$370.18 + \$80 shipping/tax reserve = **\$450.18**. With our own calibrator: \$589.18 + \$100 reserve = **\$689.18**. A recorder-specific cable/adaptor, if needed, is additional.

What could be misaligned?

Start delay. Buffering alone can add several milliseconds. One 256-sample buffer at 48 kHz lasts **5.3 ms**; two last 10.7 ms. Converter, speaker and air delays are additional. This is an example, not measured UMC latency. Air adds about **3 ms per metre** in either setup.

Trial variation. How much the delay changes between trials is **unmeasured**. Preload the sequence and use wired audio. The moment the program calls “play” is not the sound’s arrival time.

Clock drift. The interface-to-recorder mismatch is **unknown**. For scale, a hypothetical **100 ppm relative mismatch** means **6 ms per minute, or 0.36 seconds per hour**, if uncorrected. This is an illustration, not a prediction for this hardware.

How we align it

Generate a reference alongside each tone on the spare output and record it with the brain data. Measure its offset to sound arrival with the microphone. Recording the mic only on the USB interface still leaves two clocks to align. Audio output is not TTL; recorder inputs determine the adapter. A compatible neural-recorder mic input would give a more direct acoustic reference.

Timing sources: [PortAudio buffer calculation](#); [latency estimates](#); [NTi acoustic delay](#). Sound level still needs per-tone calibration at the ear.

Accurate version 1: SONIC

\$4,840.36 priced so far + TDT quotes + measurement.

Reproduce the paper's sound hardware: **USB-6361 BNC** → **SA1** → **MF1**. The TM2500C supplies GPS/PTP and a 10 MHz reference. Reuse our neural recorder; the paper's Connexus system is outside this budget.

Exact parts and buying links

One of each unless stated. USD item prices checked October 9, 2026; shipping, tax and added tariffs excluded. "Quote" means no verified current public price.

Part	Condition / source	USD
NI USB-6361 BNC, 782255-01	New / Newark	3,744.42
TimeMachines TM2500C reference	New / backorder	799.99
TDT MF1-M mono speaker kit	TDT order / quote	Quote
TDT SA1 amplifier	TDT order / quote	Quote
TDT ZB1PS powered chassis*	TDT order / quote	Quote
NI 781513-01 power supply*	New / Newark	123.42
NI 780534-01 locking USB cable*	New / DigiKey	69.75
NI 763830-01 US power cord*	New / Newark	51.00
NI 781233-02 ferrite*	New / DigiKey	38.25
Hosa BNC-58-103 output lead*	New / Pro Audio	13.53
Known-priced subtotal		4,840.36

*Reproduction additions: these exact accessories are not named by the paper. Confirm package contents to avoid duplicates. MF1 kit normally includes its stand and speaker cable/adaptor. TM2500C includes PSU, antenna and two SMB-BNC cables.

Measurement equipment is extra. The paper does not name its microphone or calibrator. Adding our page-4 measurement kit costs \$1,897.16 with a used preamp, making **\$6,737.52 known-priced**. Mic-to-DAQ, Ethernet and recorder-specific connections still need selection. MF1, SA1 and power quotes remain additional; there is **no confirmed complete price or upper limit**.

What precision does it provide?

Sound range: the paper tested **1.4–32 kHz**. MF1's stated free-field response is 1–65 kHz, raw ± 13 dB. The paper used approximately 80 dB at the ear; it gives no calibrated $\pm$ dB uncertainty.

Timing: nominal 192 kSa/s means **5.21 μ s sample spacing**. TM2500C specifies PTP below 1 μ s *plus network jitter*. Neither number establishes sound-to-neural alignment. The paper reports no quantified acoustic jitter or long-session drift.

Drift: NI's free-running 50 ppm figure equals **3 ms/min or 180 ms/hour versus ideal time**. This is not SONIC's drift after locking to GPS. Our recorder needs a compatible reference or repeated recorded markers; its relative drift is unknown. Free-running clock-only tone error is ± 0.5 Hz at 10 kHz; GPS locking changes that reference.

Primary sources: [SONIC v1](#), [Methods 6.1–6.2](#); [NI specs](#), typical at 25 °C *unless noted*; [TM2500C timing](#); [MF1 specs](#). Paper assembly 152805A-03L maps to the BNC model. An [eBay BNC offer at \\$3,498.95](#) is a cheaper lead; confirm exact SKU and accessories. SONIC recorded awake, alert sheep; Rice sedation is an unverified adaptation.

Accurate version 2: McGinley slides

\$7,551.75 priced so far + ES1 quote + measurement.

Reproduce the supplied rack: **PC** → **PXIe rack / PXI-4461** → **ED1** → **ES1**. The PXIe-6363 provides additional I/O. Our neural recorder replaces the slides' Neuropixels chain in this budget.

Exact parts and buying links

Qty 1 each. These are active asking-price leads; used calibration and function are unverified. Most rack parts are used, and several ship from China.

Part	Condition / source	USD
NI PXIe-1082 chassis	Used / eBay	2,650.00
NI PXIe-6363 I/O module	Used / eBay	2,495.36
NI PXI-4461, hybrid 186900-11L	Used / eBay	699.99
NI PXIe-8360 controller	Used / eBay	493.20
NI PCIe-8361 host card	Seller-new / eBay	197.00
NI 779500-03 MXI cable, 3 m	Seller-new / eBay	89.85
NI SHC68-68-EPM, 192061C-02, 2 m	Used / eBay	89.99
NI BNC-2090A breakout	Used / no returns	613.77
TDT ED1 + zBus + PS25F power bundle	Used / eBay NTC	209.06
TDT ES1 + matching high-voltage cable	TDT order / quote	Quote
Hosa BNC-58-103 output lead*	New / Pro Audio	13.53
Known-priced subtotal		7,551.75

*Our added output lead. ED1 bundle includes chassis/power and an incidental HB7; no separate ZB1PS is counted. Confirm ES1 cable inclusion. One SHC cable follows the diagram; extra channel banks need another. Verify actual 4461 hybrid connector and recent calibration before buying.

Measurement equipment is extra. The slides show no mic/calibrator. Our page-4 kit adds \$1,897.16, making **\$9,448.91 known-priced**. Mic/recorder wiring, any missing rack power cord and ES1 quote are additional. The host must be a compatible desktop with a free PCIe slot. Software, shipping, tax and labor are excluded.

What precision does it provide?

Sound range: ES1 is specified for **4–110 kHz**, raw ± 11 dB. It misses SONIC's 1.4–4 kHz tones. The PXI-4461's stated output flatness extends to 92.1 kHz at its maximum rate. At-ear $\pm$ dB accuracy is unknown; electrical flatness is not acoustic accuracy.

Timing: at 204.8 kSa/s, spacing is **4.88 μ s**. The DAC filter adds **156.25 μ s** at that rate; speaker and air delay are additional. The slides give no measured sound-to-neural jitter.

Drift: the 4461's internal ± 20 ppm specification corresponds to **1.2 ms/min or 72 ms/hour versus ideal time**, before aging/calibration uncertainty. Clock-only frequency error is ± 0.2 Hz at 10 kHz. Shared-rack clock settings and the recorder clock still determine relative drift; the slides do not establish either.

Sources: supplied diagram; supplied parts slide; PXI-4461 specs, pp.19,24; ES1 specs; hybrid variant. NI accuracy conditions include calibration limits; used hardware is not verified to meet them.

Accurate version 3: my recommendation

\$5,636.41 priced so far + MF2 / amplifier / power quotes.

For full 1.4–32 kHz coverage: **USB-6361** → **SA1** → **MF2-M**. Record an **M50 microphone** via **MM-1** on the DAQ and repeated timing markers on our neural recorder. This is a proposed build, not tested performance.

Exact parts and buying links

Qty 1 except batteries/resistors. Confirm actual sample rate in the driver. Main subtotal uses a new preamp. Blue names link directly to sellers or TDT's product/order route.

Part	Condition / source	USD
NI USB-6361 screw-terminal, 781442-01	New / Newark	3,030.42
TDT MF2-M mono speaker kit	TDT quote	Quote
TDT SA1 amplifier	TDT quote	Quote
TDT ZB1PS powered chassis	TDT quote	Quote
Earthworks M50 measurement mic [†]	New / eBay	1,299.00
Sound Devices MM-1 preamp [†]	New / B&H	945.00
REED R8090 calibrator [†]	New / DigiKey	219.00
Hosa HMIC-010 XLR mic cable [†]	New / Sweetwater	17.99
Gator RI-MICTP-FBM mic stand [†]	New / Sweetwater	34.99
Two Panasonic AA batteries, \$0.59 each [†]	New / Mouser	1.18
Pomona EM4970-36# output lead	New / DigiKey	23.68
Hosa PHX-106F-BULK mic output adapter	New / B&H	13.95
Two Yageo 100 kΩ bias resistors, \$0.10 each	New / DigiKey	0.20
NI 763830-01 US power cord	New / Newark	51.00
Known-priced subtotal		5,636.41

[†]Shared measurement kit: \$2,517.16 with a new preamp, or \$1,897.16 with the [\\$325 used MM-1 \(no returns\)](#). Used-preamp system subtotal: \$5,016.41. DAQ includes PSU/USB cable; M50 includes calibrator adapter; MF2 includes stand/speaker cable. Hosa's Phoenix plug needs removal and differential wiring/bias review; it is not a direct NI plug. Recorder-specific marker adapter is still unpriced. Keep preamp gain fixed, filter/limiter off; account for NI input scan timing.

What precision should we expect?

Sound: MF2's free-field band is **1–80 kHz**; M50 covers to 50 kHz. This covers SONIC's band. R8090's **±0.5 dB applies at 1 kHz**; whole-band at-ear accuracy remains unmeasured.

Timing and drift: proposed actual 200 kSa/s gives **5 μs spacing**, not 5 μs alignment. Free-running NI 50 ppm means **180 ms/hour versus ideal time**; clock-only error is ±0.5 Hz at 10 kHz. Record repeated markers and microphone arrivals to estimate offset/drift. Corrected alignment is unmeasured; **≤1 ms is a proposed acceptance goal**.

Budget decision: use the basic setup for selected audible tones; choose this version for the full SONIC band. No fixed upper budget is defensible until TDT quotes and recorder wiring are resolved. Arduino/FPGA can provide buffered playback/markers, but still needs suitable analog output, amplification, calibration and clock alignment; it is not yet a verified cheaper full-band alternative.

Specs: [MF2](#); [M50](#); [MM-1](#); [R8090](#); [NI timing](#). Our existing computer/recorder are reused; confirm NI-DAQmx host support. Software/labor, shipping, tax and added tariffs are excluded. No build or animal-ready protocol is claimed.